

Yash Uttamchandani

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EDUCATION

Year	Degree/Certificate	Institute	CPI/%
2022-2024	M.Tech/Computer Science & Engg.	Indian Institute of Technology, Kanpur	8.1/10.0
2019-2022	B.Tech/Computer Science & Engg.	Charotar University of Science & Technology, Anand	8.7/10.0
2016-2019	Diploma/Computer Engineering	A.V. Parekh Technical Institute, Rajkot	8.7/10.0
2016	GSEB(X)	Sadhu Vaswani School, Rajkot	80%

RESEARCH EXPERIENCE

- Limits on using LLM’s for predictions**(M.Tech Thesis) (June ’23 - May ’24)
Guide: **Prof. Nisheeth Srivastava**
 - Empirically demonstrated the limitations of LLMs in accurately predicting human decision-making, challenging existing claims regarding their effectiveness as cognitive models.
 - Developed and implemented a novel test-retest experimental framework across three distinct cognitive tasks to analyze and compare the predictive accuracy of LLMs to human behavior.
 - Successfully revealed the inherent inconsistencies in human decision-making over time, establishing a crucial performance benchmark for LLMs and highlighting the challenge of modeling human variability.
 - Provided evidence that LLMs, even the highly-touted CENaUR, consistently fell short of human performance in replicating human choices.

PROJECTS

- Air Pollutant Prediction using Sensor Data** (CS771) | Guide : Prof. Purushottam Kar (Aug’22 - Nov’22)
 - Built linear regression model to predict O3 and NO2 using just voltage, achieving O3 MAE 5.63 and NO2 MAE 6.3
 - Built various non-linear model by incorporating temperature, humidity, and timestamps along with voltage features
 - KNN was found best among them, achieving MAE of O3 3.1 and MAE of NO2 1.6
- Sampling and Reconstruction of Volume Data** (CS661) | Guide : Prof. Soumya Dutta (Jan’23 - April’23)
 - Implemented a Simple Random Sampling function to sample a given percentage of points from the Isabel 3D dataset
 - Performed reconstruction of volume data from the sampled points using nearest and linear interpolation methods
 - Attained a notable signal-to-noise ratio (SNR) of up to 40 for various sampling percentages

EXPERIENCE

- Interglobe Aviation (Indigo) (Data Scientist)** (June ’24 - Present)
 - Intelligent Feedback Analysis System using LLMs**
 - Engineered an incremental data ingestion pipeline from Snowflake to Databricks with daily batch jobs, handling end-to-end data cleaning and preprocessing within a production-grade pipeline.
 - Designed few-shot prompt templates to classify customer feedback into 3 sentiment classes (Positive, Neutral, Developmental) and a multi-level issue taxonomy (L1→L2→L3→L4), supporting multi-label classification where a single feedback maps to multiple categories simultaneously.
 - Conducted multi-model evaluation across Claude, LLaMA, and Mistral to benchmark classification accuracy and semantic consistency, selecting the best-performing model for production deployment.
 - Operationalized the end-to-end classification pipeline on Databricks Workflows with automated daily inference, writing structured sentiment and category outputs back to Snowflake for downstream consumption.
 - Delivered a Snowflake-built Power BI dashboard that enables business users to explore sentiment trends, category breakdowns, and dissatisfaction signals — reducing manual feedback review effort by 95%+.
 - Hotel Booking Propensity Model using Customer Behavioral Segmentation**
 - Leveraged Customer Data Platform (CDP) data containing rich behavioral features — flight frequency, booking lead time (early/late), trip purpose (business/leisure), annual trip count, and spending patterns — to identify high-propensity hotel bookers.
 - Applied K-Means clustering on the existing hotel booker population to discover 10 distinct behavioral segments, capturing diverse customer profiles and booking patterns.
 - Engineered a two-stage candidate selection mechanism — applied Euclidean distance thresholding to filter non-bookers within proximity of confirmed booker clusters, followed by cosine similarity ranking to score and prioritize users by behavioral similarity, producing a high-confidence propensity list.
 - Drove measurable business impact with a 30% increase in inactive-to-active user conversion and a 20% uplift in click-to-open rate (CTOR) of targeted hotel promotional emails and push notifications, demonstrating the effectiveness of behavioral segmentation in precision marketing.
 - LLM-Based Cargo Commodity Code Classification & Revenue Leakage Detection**
 - Identified that 20% of cargo shipment records carried mislabelled commodity codes, leading to incorrect rate application and revenue leakage; designed and delivered an end-to-end LLM-based pipeline to automatically detect and correct these misclassifications.
 - Engineered an incremental ingestion pipeline from Snowflake to Databricks ensuring daily batch runs process only new cargo bookings efficiently.

- * Built a structured prompt using priority tiers, confidence-based multi-label output, and domain-specific classification rules (e.g. special handling for high-value commodities like VALN, ICE) — leveraging both official commodity code definitions and top observed historical patterns per code as reference context.
- * Implemented a batched inference strategy (10 descriptions per LLM call) using LLaMA 4 Maverick via Databricks `ai_query()`, significantly reducing API call overhead while maintaining classification accuracy of 95% .
- * Developed a downstream rate calculation engine applying weight slab logic, commodity-specific rate overrides, and min/max price computation against actual freight market rates — directly quantifying revenue impact of each mislabelled shipment.
- * Persisted structured prediction results and freight market variance to Snowflake, powering a Power BI dashboard that enabled business and revenue teams to monitor commodity mismatches, pricing discrepancies, and potential revenue leakage.
- **ASK_PRISM — Conversational Text-to-SQL Intelligence System for Business Self-Serve Analytics**
 - * Designed and developed ASK_PRISM, an enterprise-wide conversational AI system enabling business users to query databases in natural language and retrieve insights without depending on technical or BAU teams for SQL query writing.
 - * Initiated the system with ASK_PRISM_CDP on Databricks Genie Spaces, targeting CDP data for customer and behavioral analytics; successfully deployed to production where it continues to serve business users.
 - * Architected an expansion of ASK_PRISM onto Snowflake's Cortex layer — leveraging Cortex Analyst, AI Agents, and Snowflake Intelligence — to cover enterprise-wide data domains including Sales, Air Operations (AIROPS), India Data Hub (IDH), and Catering Operations (COPS), eliminating the need to replicate data into Databricks.
 - * Configured and curated semantic models, table definitions, and domain-specific context for each data domain, enabling accurate natural language to SQL translation across diverse and complex schemas.
 - * Collaborated with a cross-functional team to roll out the system across multiple business units, democratizing data access and significantly reducing turnaround time for business queries that previously required manual SQL intervention.

ACHIEVEMENTS AND CERTIFICATIONS

- Secured **All India Rank 185** in **GATE CS 2022** amongst 77257 candidates
- **Algorithmic Toolbox:** Course Offered by University of California San Diego
- **Career Essentials in Generative AI by Microsoft and LinkedIn**

TECHNICAL SKILLS

- **Languages :** Java, Python, C++, C, SQL
- **Libraries/Utilities:** Scikit-learn, Numpy, Pandas, Seaborn, Matplotlib, AWS, Git, Flask, ~~WFLX~~
- **AI/LLM & Generative AI:** Large Language Models (LLMs), Prompt Engineering, Few-Shot Learning, Agentic AI, Multi-Agent Frameworks, Snowflake Cortex, Databricks Genie Spaces

RELEVANT COURSES

- **Postgraduate :** Introduction to Machine Learning, Big Data Visual Analytics, Program Analysis Verification and Testing
- **Undergraduate :** Data Structures & Algorithms, Operating System, Design & Analysis of Algorithms, Cloud Computing